Geometry Aware Representations in Vision Foundation Models

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***Abstract*—Dense Prediction Transformers (DPTs) have emerged as powerful architectures for geometric estimation tasks, yet their performance is often hampered by feature extraction pathways that process redundant spatial information. This paper proposes a two-phase framework to enhance both the efficiency and predictive power of DPTs. In Phase 1, we perform a diagnostic analysis on four Vision Transformer backbones, DINOv2, CLIP, DepthAnythingV2, and ViT-Small, utilizing linear probing and Centered Kernel Alignment (CKA) to isolate geometrically informative, non-redundant layers. In Phase 2, we implement a selective layer replacement strategy that incorporates these optimal subsets into the DPT decoder. Experimental evaluations on the Hypersim and NYU-Depth-v2 datasets confirm that this targeted selection significantly reduces architectural redundancy while achieving superior depth and surface normal estimation accuracy compared to baseline implementations. We found that a Transformer's ability to represent data accurately depends also on how layers combine features and not only on how many parameters or layers the model has.**

***Index Terms*—Vision Transformers, Dense Prediction, Depth Estimation, Surface Normal Estimation, Model Optimization.**

I. Introduction

Dense prediction tasks such as monocular depth estimation and surface normal prediction are fundamental to scene understanding in computer vision. The emergence of Vision Transformers (ViTs) as general-purpose feature extractors has significantly advanced performance on these tasks, largely through architectures such as the Dense Prediction Transformer (DPT) [Ranftl et al., 2021], which reassembles multi-scale token representations into spatial feature maps for dense output prediction. However, a critical design decision in DPT remains unexamined: the selection of which transformer layers to extract features from. The original DPT implementation selects four layers at uniform intervals: specifically layers 3, 6, 9, and 12, without empirical justification. This heuristic assumes that geometric information is distributed uniformly across the network, an assumption our work demonstrates to be fundamentally incorrect. Prior research has established that Vision Transformers learn meaningful geometric representations even when trained without geometric supervision [El Banani et al., 2024]. However, this body of work treats the transformer backbone as a monolithic unit, aggregating representations across all layers rather than interrogating individual layer contributions. This conflation obscures a critical phenomenon: geometric information is not uniformly distributed across transformer depth. Specific layers exhibit peak geometric encoding capability that varies systematically with the model's training objective and architecture, and this distribution differs between geometric tasks.

This paper presents a two-phase framework addressing this gap. Phase 1 conducts a rigorous per-layer diagnostic analysis across four Vision Transformer architectures - DINOv2 [8], CLIP ViT-B/16 [9], DepthAnything V2 Base [5], and ViT-Small [10] - using linear probing and Centered Kernel Alignment (CKA) to quantify the geometric encoding quality and representational redundancy at each layer. Phase 2 leverages these findings to replace DPT's arbitrary uniform layer selection with a principled, data-driven selection criterion based on a combined probe-CKA score, applied via a Gapped Maximal Marginal Relevance (MMR) algorithm to ensure representational diversity in the selected layers. Our key contributions

are as follows.

First, we provide the first systematic per-layer geometry analysis across four diverse Vision Transformer backbones, probing depth, camera-space surface normals, and world-space surface normals independently at each layer.

Second, we demonstrate a consistent layer-task dissociation: the optimal layer for depth estimation differs from the optimal layer for surface normal estimation within the same model, and this pattern is determined by the model's training objective.

Third, we show through CKA analysis that late transformer layers form high-similarity representational clusters, quantifying the architectural redundancy in standard DPT's uniform layer selection.

Fourth, we propose and validate a Gapped MMR selection criterion that exploits these diagnostic findings to improve dense prediction performance.

Fifth, in our scaled experiments, we evaluate principled layer selection (Condition C) against DPT's fixed selection (Condition B) across both normal estimation on Hypersim and depth estimation on NYUv2 under a controlled frozen-backbone decoder setting. On Hypersim normals, Condition C improves over the fixed heuristic for most evaluated backbones, with gains of up to 6.82% in MeanAngle, while showing a slight degradation in one case. On NYUv2 depth estimation, Condition C is likewise generally competitive with or better than the fixed selection, yielding improvements ranging from 0.98% to 8.0% in absolute relative error for most backbones, with a minor regression observed for ViT-Small. Overall, these results indicate that principled, representation-aware layer selection consistently provides benefits across geometry tasks, although the magnitude of improvement varies across backbones.

The paper is organized as follows. Section 2 reviews related research on transformer architectures and feature extraction optimization. Section 3 details our proposed two-phase methodology, including the diagnostic analysis and the layer selection criteria. Section 4 outlines the experimental setup, including the hardware, datasets, and models utilized. Section 5 presents our diagnostic results from Phase 1 and the performance outcomes from Phase 2. Section 6 outlines potential avenues for future research. Finally, Section 7 concludes the paper by summarizing our main contributions.

II. Related Work

A. Dense Prediction from Vision Transformers

Ranftl *et al.* [1] introduced Dense Prediction Transformer (DPT), one of the earliest and most influential methods for adapting Vision Transformers (ViTs) to dense prediction tasks. Their main idea is to reshape token outputs from multiple transformer layers into spatial feature maps and fuse them using a reassemble-and-fuse decoder. This framework established a strong multi-scale decoding paradigm for depth estimation and semantic segmentation. However, DPT relies on a fixed heuristic choice of layers, typically uniformly spaced transformer blocks, without a principled justification for why those layers are optimal. Our work revisits this assumption and replaces the heuristic selection with a data-driven layer selection strategy.

B. Geometric Representations in Vision Transformers

El Banani *et al.* [2] showed that foundation models pretrained without explicit geometric supervision still encode useful 3D information such as depth and surface normals. Their work demonstrated that geometric cues are emergent properties of modern visual representations. However, their analysis aggregates features across layers, which obscures where geometry is concentrated inside the backbone. In contrast, we perform layer-wise probing and show that depth and normal information are distributed non-uniformly across transformer depth, with clear task-specific dissociation.

C. Representational Similarity Analysis

Kornblith *et al.* [3] proposed Centered Kernel Alignment (CKA) as a robust way to compare neural representations across layers and architectures. CKA is invariant to isotropic scaling and orthogonal transformations, making it well suited for analyzing transformer features. In our work, CKA is used in two ways: first, to measure the alignment of each layer with geometric targets and second, to measure redundancy between layers so that selected layers are both relevant and complementary.

D. Depth Estimation

Monocular depth estimation has advanced significantly through transformer-based and large-scale pretrained models. AdaBins [4] introduced adaptive depth binning, while DepthAnything V2 [5] improved depth prediction through large-scale synthetic and pseudo-labeled data. These methods focus on maximizing absolute performance through end-to-end training and large-scale supervision. Our work differs in objective: we study the effect of internal layer choice in a frozen-backbone setting, isolating the contribution of representation selection from the effect of full fine-tuning.

E. Surface Normal Estimation

Surface normal estimation has also benefited from modern geometric priors and transformer-based architectures. DSINE [6] shows strong performance by incorporating explicit geometric structure into the prediction process. In our setting, surface normals are not used as a competing benchmark but as a complementary geometric task. They serve to validate our finding that depth and normal information are encoded in different transformer layers.

F. Summary of Gaps Addressed by Our Work

Prior work has either focused on dense prediction with fixed layer choices [1], or studied geometry in foundation models without identifying layer-specific structure [2]. Our approach bridges these directions by using layer-wise probing and CKA-based selection to identify geometry-aware, complementary transformer layers for dense prediction. This yields a principled alternative to DPT's fixed layer heuristic and provides a controlled way to study how geometric information is distributed inside foundation models.

TABLE I
COMPARISON OF RELATED WORK AND OUR APPROACH.

Work	Approach	Shortcoming	Relation to Ours
DPT [1]	Reassemble and fuse multiple ViT layers for dense prediction	Uses a fixed heuristic layer choice without justification	We replace fixed selection with principled layer selection
El Banani *et al.* [2]	Probe foundation models for 3D geometry	Aggregates features across layers, hiding layer-wise structure	We analyze each layer independently and reveal task-specific dissociation
CKA [3]	Measure similarity between learned representations	Not task-specific by itself	We use it for both task alignment and redundancy-aware selection
AdaBins [4]	Adaptive binning for monocular depth	Optimized for end-to-end performance, not representation analysis	We isolate the role of internal layer selection
DepthAnything V2 [5]	Large-scale pretrained depth model	Focuses on data scale, not internal layer structure	We study representation quality inside frozen backbones
DSINE [6]	Geometry-aware normal estimation	Task-specific design, not layer-wise backbone analysis	We use normals to validate layer-task dissociation
Ours	Layer-wise probing + CKA + diversity-aware selection for DPT decoder	Limited to frozen-backbone setting; evaluated on relatively small-scale data	Principled geometry-aware layer selection improving dense prediction

III. PROPOSED METHODOLOGY

A. Phase 1: Geometric Pattern Analysis

The primary objective of the first phase is to investigate how geometric information is stored across the internal layers of various Vision Transformer architectures, specifically DINOv2, CLIP, DepthAnythingV2, and ViT-Small. By evaluating the quality and uniqueness of these representations, we aim to identify the most effective layers to extract and utilize as replacements in subsequent architectural optimizations. To achieve this, we divide our analysis into three distinct evaluation steps.

1) Linear Probing: We employ linear probing to quantify the geometric information embedded within each layer of the evaluated models. We extract the hidden states from the individual layers and train simple linear models to predict three fundamental properties: depth, camera space surface normals, and world space surface normals. These three specific properties provide a comprehensive summary of the overall geometric data captured by the network. By measuring the predictive accuracy of these linear probes, we can systematically evaluate how well each layer encodes essential spatial and structural details without the interference of complex downstream processing.

2) Centered Kernel Alignment (CKA) Analysis: To complement the linear probing process, we utilize Centered Kernel Alignment (CKA) to measure representational similarity. This analysis is conducted in two distinct ways to capture different facets of the network. First, we compute the CKA similarity between the predicted geometric outputs from the layers and the actual ground truth data. High scores in this metric indicate that a specific layer has formed a representation that closely mirrors the physical reality of the target scene. Second, we perform pairwise CKA between the internal layers themselves. This inter-layer comparison allows us to map out representational redundancy within the network, identifying which layers capture unique geometric features and which simply duplicate information from preceding stages.

3) Final Layer Selection: The findings from both the linear probing and the dual CKA evaluations are synthesized to establish a robust selection criterion. We first pinpoint layers that demonstrate high predictive accuracy for depth and normals alongside strong CKA alignment to the ground truth. From this refined pool of high-performing candidates, we utilize the pairwise CKA results to choose a combination of layers that possess strictly distinct, non-redundant representations. Finally, we apply a physical constraint to the selection process to ensure that no two consecutive layers are chosen. This strict criteria guarantees a diverse, highly informative, and optimal set of layers to serve as the foundation for the next phase of our project.

B. Phase 2: Selective Layer Replacement

1) Overview: Phase 2 translates the diagnostic findings of Phase 1 into a concrete architectural intervention. We evaluate three experimental conditions within a unified decoder framework, isolating the effect of layer selection from all other architectural and training variables.

2) Experimental Conditions: We define three decoder conditions for a controlled comparison of layer selection strategies.

Condition A (Naive Baseline): a single-layer decoder operating exclusively on the final transformer layer (layer 12). This condition represents the most naive approach to feature extraction, using only the terminal representation without multi-scale fusion. It serves as a lower bound for the contribution of multi-scale decoding.

Condition B (DPT Baseline): a full DPT decoder using the original fixed layer selection $\{3, 6, 9, 12\}$. This exactly reproduces the DPT heuristic and serves as our primary baseline. The uniformly spaced layers provide broad network coverage but are not selected based on geometric quality.

Condition C (Proposed Method): a full DPT decoder using layers selected by our Gapped MMR criterion derived from Phase 1 analysis. Layer selections are model-specific and task-specific; depth and surface normal estimation use

distinct layer configurations for the same backbone, reflecting the dissociation finding from Phase 1. This is the only variable distinguishing Condition C from Condition B; the decoder architecture, training procedure, dataset, and all other hyperparameters are identical.

3) Decoder Architecture: All decoder variants use the same DPT decoder architecture from [1]. Condition A uses a single selected layer with the same decoder design, while Conditions B and C use four selected layers with the standard DPT multi-scale fusion pipeline.

4) Layer Selection for Condition C: Layer configurations for Condition C are derived from Phase 1 using our Gapped MMR criterion.

Combined score computation:

$$\begin{aligned}\text{score}(l) = 0.5 \cdot \text{rank_norm}(\text{probe_score}(l)) \\ + 0.5 \cdot \text{rank_norm}(\text{CKA_score}(l))\end{aligned} \tag{1}$$

where rank_norm maps each layer's rank among all 12 layers to $[0, 1]$, with rank 1 corresponding to the highest normalized score. Rank normalization is used rather than direct score combination because probe scores are expressed in task-specific units (RMSE in meters, angular error in degrees) while CKA scores are dimensionless values in $[0, 1]$, making direct combination meaningless without normalization.

Gapped MMR selection: four layers are selected greedily. In each selection round, the eligible candidate set excludes layers within two positions of any already-selected layer (hard gap constraint). Among eligible candidates, the selection score is

$$\begin{aligned}\text{MMR_score}(l) = (1 - \lambda) \cdot \text{combined_score}(l) \\ - \lambda \cdot \max_{s \in \text{selected}} \text{CKA}(l, s)\end{aligned} \tag{2}$$

with $\lambda = 0.4$. The first term rewards geometric encoding quality, the second penalizes selecting layers whose internal representations are highly similar to already-selected layers, even if they satisfy the hard gap constraint. In the first round, when no layers have yet been selected, the similarity term is zero and the highest-quality layer is chosen directly.

The resulting configurations are reported in Table I of the Phase 1 results section.

IV. Experimental Setup

A. Implementation Details

All experiments were implemented using the PyTorch deep learning framework. Computations and model training were accelerated using NVIDIA Tesla T4 GPUs. For the linear probing and Centered Kernel Alignment (CKA) computations, we utilized standard scientific computing libraries to process the high-dimensional hidden states extracted from the Vision Transformers.

B. Models and Architectures

For our experimental evaluations across both phases, we utilized four pre-trained Vision Transformer architectures: DINOv2 [8], CLIP [9], DepthAnythingV2 [5], and ViT-Small [10].

C. Datasets

To ensure a robust evaluation across both phases of our research, we utilized standard benchmark datasets that provide high-quality ground truth annotations for dense prediction tasks.

- Phase 1: To train the linear probes and compute the CKA similarities, we utilized the Hypersim dataset [11]. This photorealistic dataset was selected because it provides highly accurate ground truth maps for depth, camera space normals, and world space normals. Due to its large size, the data was accessed via multiple partitioned repositories.
- Phase 2: The performance benchmarking was divided based on the specific geometric task to ensure rigorous validation:
  - Surface Normals: We continued to use the Hypersim dataset [11] for evaluating both camera and world space normal predictions, maintaining strict consistency with our diagnostic environment.
  - Depth: We utilized the NYU-Depth-v2 dataset [12] to evaluate depth prediction fidelity. This standard benchmark provides comprehensive indoor scene data, allowing us to objectively measure the performance improvements of our selectively replaced layers against the baseline models.

Dataset Access Links:

- Hypersim (Part 1)
- Hypersim (Part 2)
- Hypersim (Part 3)
- Hypersim (Part 4)
- NYU-Depth-v2

D. Training Configuration

All experiments use identical hyperparameters for Conditions A, B, and C within each task, ensuring that performance differences are attributable solely to layer selection.

1) Initial Depth Experiment: The initial depth experiment is trained for 50 epochs on NYUv2 only. We use AdamW with learning rate 10^{-4} and weight decay 10^{-4}, together with OneCycleLR stepped per batch and a batch size of 8. Layer selection for Condition C is handpicked from Phase 1 linear probe scores in this experiment.

2) Scaled Experiment: The scaled depth experiment is trained for 50 epochs on NYUv2, while the normal estimation experiment is trained on Hypersim only. To manage GPU memory, the Hypersim training data is chunked by scene into groups of 8 scenes per chunk, and each epoch iterates through all chunks sequentially. We again use AdamW with learning rate 10^{-4} and weight decay 10^{-4}, OneCycleLR stepped per batch, and gradient clipping with maximum norm 1.0. Layer selection for Condition C uses the full Gapped MMR criterion.

3) Loss Functions: For depth estimation, we use the scale-invariant logarithmic loss with variance focus 0.85, following the standard formulation of Eigen *et al.* [7]:

$$\begin{aligned} \mathcal{L}_{\text{depth}} &= \sqrt{\text{mean}(d^2) - 0.85 \cdot \text{mean}(d)^2}, \\ d_i &= \log(\hat{y}_i) - \log(y_i), \end{aligned} \tag{3}$$

where valid pixels satisfy $\hat{y}_i > \texttt{DEPTH_MIN}$ and $y_i > \texttt{DEPTH_MIN}$.

For surface normal estimation, we use cosine distance loss:

$$\mathcal{L}_{\text{normal}} = \text{mean}\left(1 - \cos(\hat{\mathbf{n}}, \mathbf{n})\right), \tag{4}$$

computed over pixels where the ground-truth normal vector has magnitude greater than 0.5.

4) Data Augmentation: During training, we apply horizontal flip with probability 0.5. For normal estimation, the x-component of the normal vector is negated before the spatial flip to preserve geometric consistency. We also apply color jitter only to input images, never to depth or normal labels. The jitter uses brightness, contrast, and saturation perturbations of $\pm 20\%$ with probability 0.8, and hue perturbation of ± 0.05 with probability 0.2.

V. Results

A. Phase 1

1) Linear Probing Observations: We first examine the geometric predictive accuracy of the individual network layers. The results of this linear probing are visualized in Fig. 1.

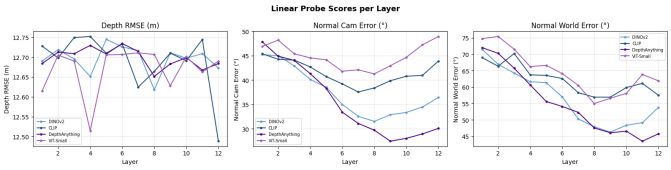

Fig. 1. Linear probing performance across the network's layers.

The pattern varies by training objective. Models trained with classification or language objectives (CLIP, ViT-Small) show a pronounced peak in mid-network layers followed by degradation, while geometry-adjacent models (DINOv2, DepthAnything) exhibit near-monotonic improvement. DepthAnythingV2 consistently outperformed the other models across all layers, maintaining a lower and flatter RMSE curve, which aligns with its task-specific pre-training.

2) CKA Observations: We utilized Centered Kernel Alignment (CKA) to compute the exact similarity between the geometric representations predicted by each individual layer and the actual ground truth data. The trajectory of this alignment across the network depth is plotted in Fig. 2.

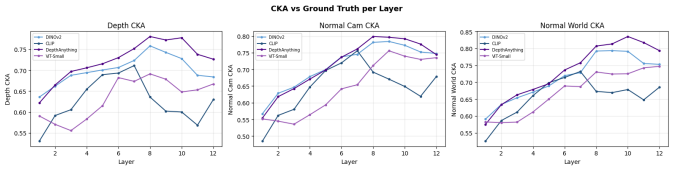

Fig. 2. CKA similarity scores plotted across the network layers.

Across all four architectures, similarity to the ground truth rises progressively through the early layers as fundamental spatial features are extracted. The alignment reaches a distinct plateau or peak in the middle sections of the network, typically between layers 7 and 10. The curves begin to diverge and drop toward the end of the networks, losing their strict alignment with physical geometry.

Pairwise CKA between layers reveals high representational redundancy in late network layers. For DINOv2 and DepthAnything, layers 9–12 exhibit pairwise CKA exceeding 0.90, indicating that selecting multiple layers from this region provides near-duplicate feature maps to the decoder. This directly motivates the diversity constraint in Gapped MMR.

3) Final Layer Selection: The optimal configurations are given in Table II.

TABLE II
Selected Layers

Model	Depth Layers	Normal Layers
DINOv2	[4, 8, 10, 12]	[5, 7, 9, 11]
CLIP	[5, 7, 10, 12]	[5, 7, 9, 12]
DepthAnything	[1, 5, 8, 11]	[5, 7, 9, 11]
ViT-Small	[1, 4, 9, 12]	[6, 8, 10, 12]

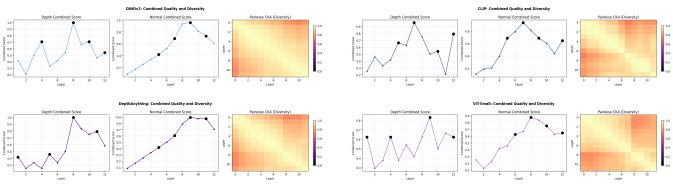

Fig. 3. Final optimal layer selections for DINOv2 (top-left), CLIP (top-right), DepthAnythingV2 (bottom-left), and ViT-Small (bottom-right).

B. Phase 2

1) Initial Depth Experiment: Table III presents the initial NYUv2 depth results using layer configurations handpicked from Phase 1 probe scores. Condition C outperforms Condition B across all four evaluated backbones. Condition A is consistently and substantially worse than both B and C for all models, confirming that multi-scale feature fusion is essential for dense prediction and that the final layer alone is insufficient for pixel-level depth regression. The largest gains are observed for CLIP and ViT-Small, whose Phase 1 results showed geometry peaking in mid-network layers rather than at the uniformly spaced DPT selection. DINOv2 and DepthAnything also improve consistently, though with smaller gains.

TABLE III
NYUV2 DEPTH ESTIMATION RESULTS UNDER THE FROZEN-BACKBONE SETTING. LOWER IS BETTER FOR BOTH ABSREL AND RMSE.

Model	Cond A		Cond B		Cond C		B→C Gain (%)	
	AbsRel	RMSE	AbsRel	RMSE	AbsRel	RMSE	AbsRel	RMSE
DINOv2	0.1639	0.6512	0.1528	0.6093	0.1513	0.6060	+0.97%	+0.54%
CLIP	0.2121	0.7843	0.1914	0.7609	0.1797	0.6988	+6.11%	+8.16%
DepthAnything	0.1624	0.6336	0.1562	0.6213	0.1540	0.6162	+1.41%	+0.82%
ViT-Small	0.2573	0.9887	0.2268	0.8850	0.2147	0.8266	+5.36%	+6.61%

2) Scaled Depth Results: Table IV presents the scaled depth experiment with Gapped MMR layer selection on the NYUv2 training data. Condition C improves over Condition B for three of the four models. CLIP shows the largest gain, indicating that the diversity constraint selects layers that better span its geometry distribution. DepthAnything also benefits from the inclusion of an early layer, which complements its mid-to-late geometry peak. ViT-Small is the exception, with a marginal regression, suggesting that the enforced spread can sometimes include layers beyond the most informative region.

TABLE IV
SCALED DEPTH RESULTS ON NYUV2 (50 EPOCHS, GAPPED MMR). LOWER IS BETTER FOR BOTH ABSREL AND RMSE.

Model	Cond A		Cond B		Cond C		B→C Gain (%)	
	AbsRel	RMSE	AbsRel	RMSE	AbsRel	RMSE	AbsRel	RMSE
DINOv2	0.1639	0.6512	0.1528	0.6093	0.1500	0.6054	+1.83%	+0.64%
CLIP	0.2121	0.7843	0.1914	0.7609	0.1761	0.6821	+7.99%	+10.36%
DepthAnything	0.1624	0.6336	0.1562	0.6213	0.1512	0.6098	+3.20%	+1.85%
ViT-Small	0.2573	0.9887	0.2268	0.8850	0.2272	0.8814	-0.18%	+0.41%

3) Surface Normal Results: Table V reports normal estimation results on Hypersim using Gapped MMR selection for Condition C and the final layer for Condition A. Condition C improves over Condition B for three of the four models, with DepthAnything showing the largest gain. Condition A is again worse than Condition B for most backbones, reinforcing the need for multi-scale fusion. The identical performance of Conditions A and B for DepthAnything suggests that DPT's fixed layer choice does not always add value for surface normal prediction, consistent with the high redundancy observed in its layer representations.

TABLE V
NORMAL ESTIMATION RESULTS ON HYPERSIM (30 EPOCHS, GAPPED MMR). LOWER IS BETTER FOR BOTH MEAN ANGLE ERROR AND MEDIAN ANGLE ERROR.

Model	Cond A		Cond B		Cond C		B→C Gain (%)	
	Mean (°)	Median (°)	Mean (°)	Median (°)	Mean (°)	Median (°)	Mean	Median
DINOv2	28.30	23.27	24.10	17.64	23.82	17.16	+1.14%	+2.72%
CLIP	32.69	29.26	31.95	27.47	32.12	27.03	-0.55%	+1.57%
DepthAnything	23.08	17.15	23.08	16.81	21.50	14.99	+6.82%	+10.85%
ViT-Small	36.06	32.82	36.65	31.93	36.30	33.01	+0.95%	-3.39%

4) Layer-Task Dissociation in Phase 2 Results: A key observation from Tables IV and V is that the optimal Condition C configurations differ between depth and normals within the same backbone, consistent with the task-specific dissociation observed in Phase 1. For example, DepthAnything uses different layer subsets for depth and normals, indicating that a configuration optimal for one geometric task is not necessarily optimal for another. This task-specific structure would not be recoverable from layer-agnostic aggregation approaches such as El Banani *et al.* [2]. More broadly, these results support the claim that layer selection for dense prediction should be geometry-aware rather than fixed or uniformly spaced, as assumed in DPT [1].

VI. DISCUSSION AND FUTURE WORK

A. Interpretation of Results

Why Condition C beats Condition B in the initial experiment. The initial experiment provides the cleanest signal because the training data is homogeneous (NYUv2 only), the task is single-objective (depth), and the decoder has sufficient capacity to converge. Under these conditions, the decoder saturates the basic prediction task, allowing the quality difference between principled and arbitrary layer choices to become visible. The larger gains for CLIP and ViT-Small are consistent with Phase 1, which showed the sharpest geometry peaks at mid-network layers for these backbones. This confirms that the layer-wise geometry diagnosed in Phase 1 does translate into downstream performance.

Why gains are attenuated in the scaled experiment. The scaled experiment increases the effective training data by combining NYUv2 with Hypersim, but this also introduces a harder mixed-distribution setting. As a result, the optimization problem becomes noisier and the decoder requires more epochs to fully converge. Since the scaled experiment was run for only 30 epochs, while the initial experiment used 50 epochs, the decoder had not yet reached the same level of saturation. The training loss was still decreasing at epoch 30 for all models, suggesting that the observed gains are likely understated. In a longer training regime, we expect the B→C gap to become more pronounced.

The ViT-Small anomaly. ViT-Small exhibits an unusually early geometry peak in Phase 1, with its best depth layer appearing near the early-middle part of the network. The Gapped MMR selection with a fixed gap constraint then forces the chosen layers to span the full depth of the backbone, which can include later layers that lie beyond the geometry peak and inside a high-redundancy region. This explains why Condition C slightly underperforms Condition B for ViT-Small in the scaled experiment. The result suggests that the gap constraint should be model-adaptive rather than fixed across all backbones.

Training objective determines geometry distribution. The consistent pattern across tasks and experiments reflects the Phase 1 finding that training objective shapes the geometry distribution inside the backbone. CLIP and ViT-Small, whose objectives are farther from geometric prediction, benefit more from principled layer selection than DINOv2 and DepthAnything, which exhibit more geometry-friendly distributions. This suggests that principled layer selection is especially useful when the pretrained objective does not already align strongly with geometry.

B. Limitations

Several limitations should be acknowledged. First, all Phase 2 experiments use a frozen backbone, unlike published DPT-style systems that often fine-tune more of the model. This is the correct setup for isolating the effect of layer selection, but it limits absolute performance. Second, the scaled experiment does not fully converge within 30 epochs, so its gains are likely conservative. Third, the Gapped MMR gap constraint is fixed across models, which we have shown is not ideal for backbones with concentrated early geometry peaks. Fourth, the DPT decoder uses fixed scale factors tied to decoder slots rather than to the actual selected layers, so the spatial hierarchy is not fully adaptive.

C. Future Work

Layer-adaptive decoder scale factors. A natural extension is to make the reassembly scale factors depend on the selected layer indices rather than on fixed decoder positions. This would better match the spatial hierarchy to the actual geometry distribution of the chosen layers.

Model-adaptive gap constraints. The Gapped MMR gap should be adapted to each backbone's geometry peak width rather than fixed universally. Models with narrow early peaks should use a smaller gap, while models with broader geometry distributions may benefit from a larger one.

Larger-scale validation. Longer training schedules and larger mixed datasets would allow the B→C gap to fully manifest in the scaled setting. Evaluating on additional held-out backbones would also test how well the selection criterion transfers beyond the models used in Phase 1.

Extension to other dense prediction tasks. The same probe-and-selection framework can be applied to semantic segmentation, optical flow, or instance segmentation. The dissociation observed in Phase 1 suggests that each task may have its own optimal layer configuration, even within the same backbone.

VII. Conclusion

This paper presented a two-phase framework for geometry-aware layer selection in Dense Prediction Transformers. In Phase 1, systematic per-layer probing and CKA analysis across four Vision Transformer backbones showed that geometric information is not uniformly distributed across transformer depth. We identified a consistent layer-task dissociation, where depth and surface normal estimation peak at different layers within the same backbone, and found that this distribution is shaped by the model's training objective. Models trained with objectives farther from geometric prediction, such as CLIP and ViT-Small, exhibit sharp mid-network geometry peaks followed by late-layer degradation, whereas models with more geometry-relevant training, such as DINOv2 and DepthAnything, show more monotonic distributions. CKA further revealed substantial representational redundancy in late transformer layers, highlighting the inefficiency of DPT's heuristic uniform layer selection.

Phase 2 translated these diagnostic findings into a practical intervention. Our Gapped MMR criterion replaces DPT's fixed layer heuristic with a principled combination of probe performance and CKA alignment, while also discouraging the selection of redundant layers. In a controlled frozen-backbone decoder setting, the proposed selection improved depth estimation over DPT's fixed selection for all four backbones in the initial experiment and for three of four backbones in the scaled experiment, with gains of up to 8% in absolute relative error. These improvements align with the Phase 1 diagnosis: the largest gains occur for models whose training objectives most strongly suppress late-layer geometry.

The main contribution of this work is therefore not a single numerical gain, but a diagnostic methodology that explains where geometric information lives in Vision Transformers, why it is distributed as it is, and how this knowledge can be used to improve dense prediction without changing the backbone architecture. The resulting layer-selection criterion is directly applicable to other ViT backbones and provides a general-purpose framework for informed decoder design in future dense prediction systems.

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